

Projects

Dino jump

the-cc.io/dino-jump-editor

Build a **Dino Jump** game where **Pico** leaps over moving obstacles and scores a point for every one they avoid.

Press **space** to jump!



1 Add a character

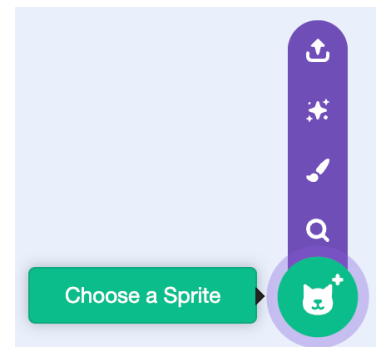


- ☐ Go to **the-cc.io/dino-jump-editor**. The starter project opens with an empty Stage ready for your game.

Click **Choose a Sprite**.

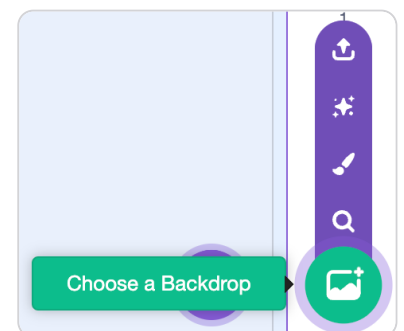
Find **Pico** in the sprite library and click on it.

- ☐ **Check your project** **Pico** appears on the Stage. There's no code to run yet.



2 Choose a backdrop

- ☐ Click **Choose a Backdrop**. Find **Desert** in the backdrop library and click on it.



- ☐ **Check your project** Your Stage now shows **Pico** in the desert.

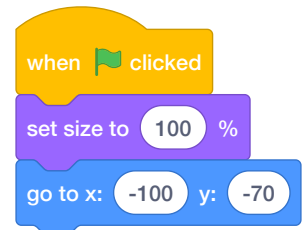
Pico might be in a different place on your Stage. You'll set its starting position next.



3 Set the starting position



- ☐ On **Pico**, add a script to set its size and place it near the bottom-left of the Stage.



- ☐ **Now run your code** Click the green flag. Drag **Pico** somewhere else, then click the green flag again.

Pico returns to x: -100, y: -70 at 100% size.

4 Jump and land

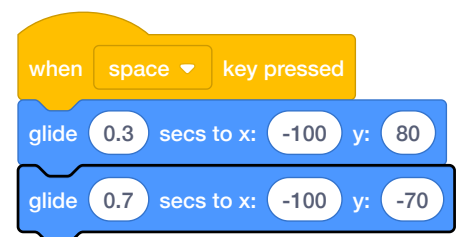


- ☐ On **Pico**, start a new script with a **when space key pressed** block. Add a **glide () secs to x: () y: ()** block with these values.



- ☐ On **Pico**, add another **glide () secs to x: () y: ()** block to the bottom of the space-key script.

Use the same landing position as the green-flag script.



- ☐ **Now run your code** Click the green flag, then tap the space bar once. **Pico** jumps up and lands at y: -70.

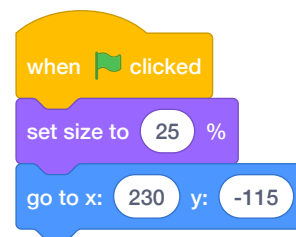
Wait for **Pico** to land before tapping space again.

5 Add Dinosaur5



- ☐ Click **Choose a Sprite**, find **Dinosaur5** and add it to your project.

- ☐ On **Dinosaur5**, add this script to shrink it and put it near the right edge of the Stage.



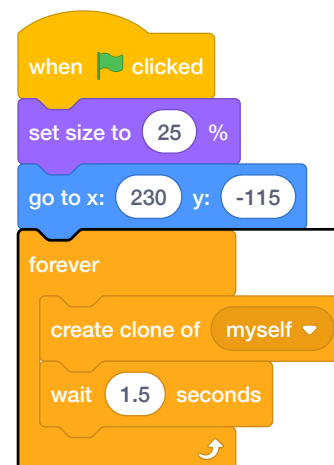
- ☐ **Now run your code** Click the green flag. **Dinosaur5** should be much smaller than **Pico**, with its feet at about the same height.

If it appears near **Pico**'s head, check that its size is **25%** and its **y** position is **-115**.

6 Create Dinosaur5 clones



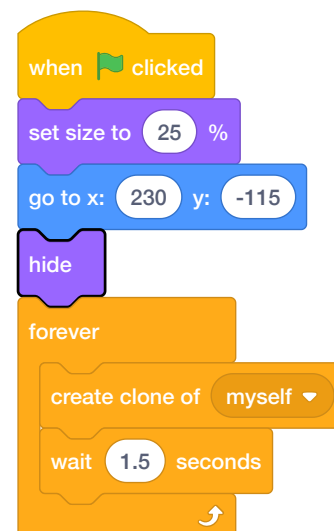
- ☐ On **Dinosaur5**, add a **forever** loop to the setup script. Inside it, create a clone and wait **1.5** seconds before making the next one.



- ☐ Add **hide** before the loop to hide the original **Dinosaur5**.

Tip A clone is a copy of a sprite. These clones start hidden because they copy the hidden **Dinosaur5**.

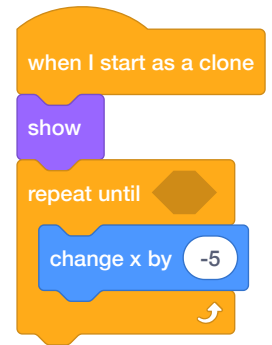
- ☐ **Now run your code** Click the green flag. **Dinosaur5** disappears. The clones are being created, but you'll make them visible in the next step.



7 Move and remove the clones



- On **Dinosaur5**, start a new script with **when I start as a clone** and **show**. Add a **repeat until** loop. Inside it, add **change x by ()** and set it to **-5**. Leave the condition empty for now.



- From **Operators**, drag a **less than** block into the loop's condition. Put **-230** in its right-hand slot.

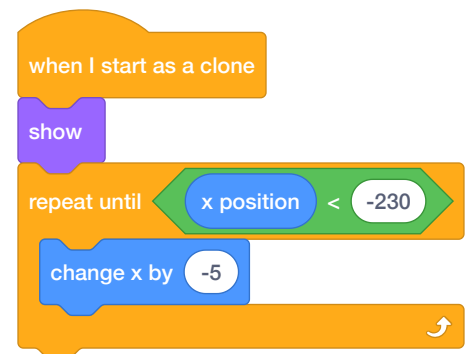


From **Motion**, put an **x position** block in the left-hand slot.



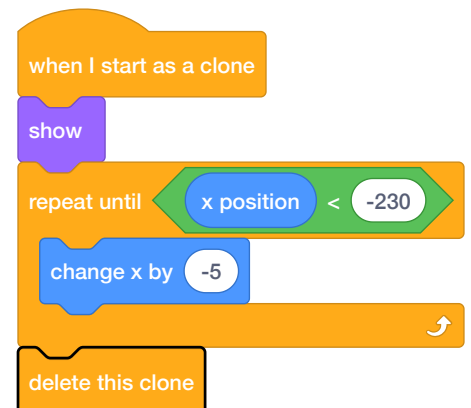
The finished script looks like this:

Tip A negative number in **change x by ()** moves the clone to the left.



- On **Dinosaur5**, add **delete this clone** below the movement loop, outside it.

Tip Use **-230**, not **-240**. Scratch keeps a little of each sprite on the Stage, so a small clone can get stuck before its **x position** reaches **-240**.

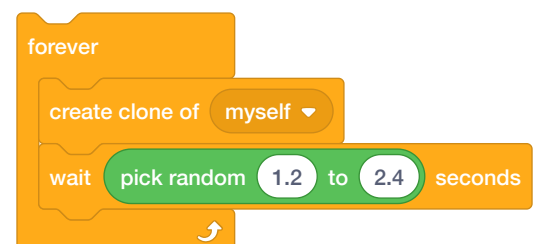


- Now run your code** Click the green flag. Watch several **Dinosaur5** clones cross the Stage. Each should disappear near the left edge, with none left behind.

8 Randomise the gaps



- In **Dinosaur5**'s **forever** loop, put a **pick random () to ()** block inside **wait () seconds**. Use **1.2** and **2.4** to leave enough time between clones for **Pico** to land.



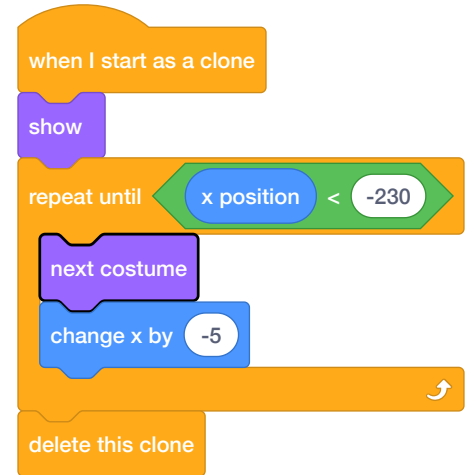
- ☐ **Now run your code** Click the green flag and watch several **Dinosaur5** clones appear. The gaps should vary. Tap space to practise jumping over one clone at a time.

9 Animate the clones



- ☐ In **Dinosaur5**'s clone script, add **next costume** inside the movement loop, before **change x by ()**.

- ☐ **Now run your code** Click the green flag. Each **Dinosaur5** clone changes costume as it moves and disappears near the left edge.
- If any clones stay there and wiggle, check that the loop uses **-230** and that **delete this clone** is below it.

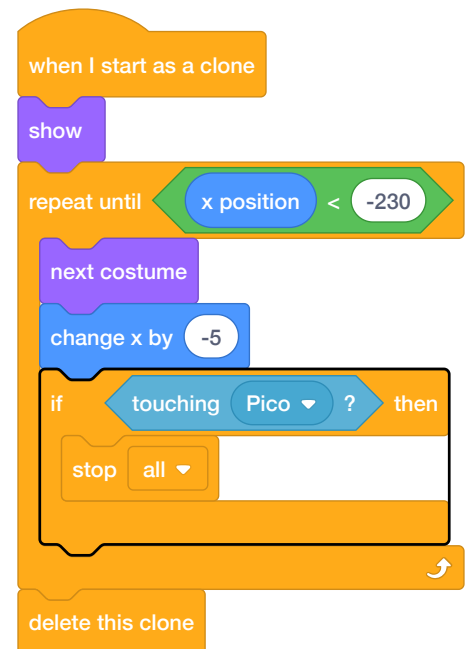


10 Stop after a collision

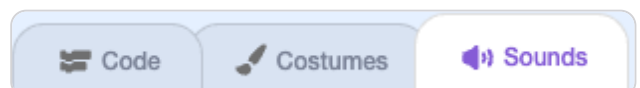


- ☐ In **Dinosaur5**'s movement loop, add an **if then** block with a **touching ()** condition.
- Choose **Pico** in the dropdown. Put **stop all** inside the **if** block.

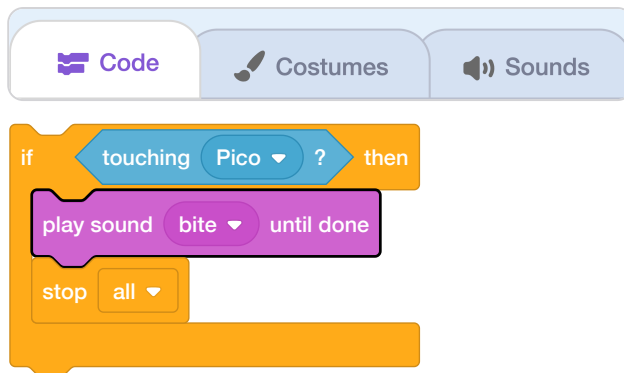
Tip The **touching ()** block starts with **mouse-pointer** selected. Click its dropdown arrow and change this to **Pico**.



- ☐ With **Dinosaur5** selected, open the **Sounds** tab. Select **bite** and click the play button to hear it.



- ☐ Return to the **Code** tab. Inside the collision check, add **play sound () until done** before **stop all**. Choose **bite** in the sound dropdown.



- ☐ **Now run your code** Click the green flag and let a **Dinosaur5** clone catch **Pico**. **bite** plays until it finishes, then the game stops.

11 Add a score



- ☐ With **Dinosaur5** selected, make a variable called **score**. Choose **For all sprites**.

Variables

Make a Variable

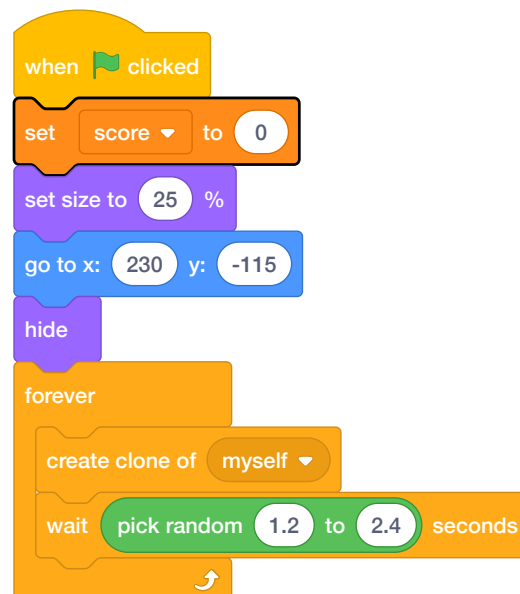


score

Keep its checkbox ticked so it appears on the Stage.

- ☐ In **Dinosaur5**'s setup script, set **score** to **0** when the green flag is clicked.

- ☐ **Now run your code** Click the green flag. The score appears on the Stage and resets to **0**. It won't go up until you add the scoring block next.



12 Score each successful jump



- ☐ In **Dinosaur5**'s clone script, add **change score by ()** after the movement loop and **before delete this clone**.

Tip

The score only changes when a **Dinosaur5** clone reaches the left edge without touching **Pico**. It goes up a little after the jump, when the clone disappears.

- ☐ **Now run your code** Click the green flag and jump over a clone. Watch it reach the left edge: **score** should go up by **1**.

Restart and let a clone hit **Pico**. You shouldn't earn a point for that clone.



13 Challenge



Choose one idea to try:

- 1 Change the speed in **Dinosaur5**'s clone script using **change x by ()**. Try **-7** for faster clones or **-4** for slower ones.
- 2 Change the gaps using **pick random () to ()** in **Dinosaur5**'s setup script. Larger numbers leave wider gaps. Keep the shortest gap at least **1.2** seconds while you practise, so **Pico** has time to land.
- 3 Replace **Pico**, **Dinosaur5** or **Desert** to give your game a new theme.
- 4 Change the glide times or height of **Pico**'s jump.
- 5 Move **Dinosaur5** higher or lower by changing its starting **y** position.
- 6 Make the clones move faster as **score** increases.
- 7 Add a sound or costume animation when **Pico** jumps.
- 8 Add a game-over backdrop before the scripts stop.

- ☐ **Try your change** Change one thing at a time and play a few rounds. Check that your idea works and that **Pico** can still jump over the clones and land between them.

Remember to save your project: open **Save & download** (the disk icon on the left of the Code Editor), then **Log in to save** or click **Download project**.